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# Transforming Health Insurance Claims Adjudication with Blockchain-based Solutions

Wael El-Samad*, Mirna Atieh[a], Mehdi Adda[b]

*aBusiness Computer Department, Faculty of Economic Sciences and Administration Lebanese University Hadat, Lebanon*
*bDépartement de mathématiques, informatique et génie, Université du Québec à Rimouski, 300 Allée des Ursulines, QC G5L 3A1, Canada*

**Abstract**

Over the past decade, the blockchain technology has gained a significant traction with the introduction of the "bitcoin", the first cryptocurrency back in 2008. The rapid growth of "bitcoin" as a currency and the Bitcoin as a protocol has served as a wake-up call to the IT community which began to realize the real potential of blockchain technology. Blockchain technology is capable of filling many gaps that traditional IT solutions struggle to address. In the other hand the increasing complexity of adjudicating health insurance claims is an urge to introduce more secure and efficient claims adjudication system. Blockchain technology has the potential to reshape the claims adjudication process not only due to its decentralized and tamper-proof features but to its consensus protocols and much more other characteristics. This article studies some of the most recent researches on blockchain solution for health insurance claim adjudication, evaluates the its feasibility and identifies areas where blockchain technology could be applied to improve efficiency and accuracy. This study will also examine the potential benefits and challenges of implementing a blockchain-based system, like reducing administrative costs and improving fraud detection.

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*Keywords:* Blockchain; Claim adjudication; consensus model; distributed ledger; Smart contracts.

## 1.Introduction

The following literature review focuses on a proposed blockchain solution for the health insurance claims processing and adjudication system only. Claims for other insurance sectors are excluded.

* Corresponding author. Tel.: +1-647-226-0772
*E-mail address:* welsamad@gmail.com

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The fact that the insurance industry faces various challenges, particularly in countries where automated IT systems are still lacking, resulting a lengthy claims settlement process, taking anywhere from a few days to a few weeks, or even months [8]; In the other side of the spectrum, third-party companies are playing the role of claims processing facilitator for the economies with advanced IT claims adjudication systems, these central claims adjudication entities are often content with the revenue generated by their automated systems, making it difficult to introduce to the market new technologies, but for these companies the threat of disruption is only a matter of time. Blockchain technology presents an opportunity to address these challenges by reducing claim processing cost and time, eliminating the need for central organizations, and improving security and privacy through its shared database, consensus protocols and cryptographic. This helps to reduce fraud and prevent system abuse, which is common in current systems.

The main objective of this article is to collect and analyze selected studies that have been done recently (in the last two to three years) that explore the use of blockchain technology for healthcare claim adjudication. While examining the different blockchain approaches employed in each study, we acknowledge that some of these articles provide a complete solution and detail the arguments to back up the claim that Blockchain presents a solid alternative of the existing system, However, it should be noted that not all studies offer this level of support. In this article we are not going to detail these results, we have a touch base of that by the end of this paper. It is important to mention as well, that we did not plan to introduce or propose our own blockchain solution; this decision was based on two main reasons, firstly: such approach requires a detailed study spanning into multiple sections and chapters, which is not suitable for the scope this paper; and Secondly, our goal in this study is raise awareness among insurance industry stakeholders regarding the potential benefit of such solutions. This is why, this article provides a high-level overview and discussion rather than an in-depth technical exploration.

First, we will start with a technical fundamental overview of the blockchain technology and its key features in part 2, in part 3 we will compare the workflow between the current claim adjudication system and a blockchain one. We will then discuss in part 4 the results of an extensive literature review, covering the most recent and relevant articles incorporating the use of blockchain in health insurance claims processing.

Our approach is to 1-collect form these articles all the challenges facing the in-place claim adjudication system, then 2-we will present the proposed blockchain solution of each article, 3-and we will as will highlight specifically how each study aimed to tackle down on the fraud issue, and this due to its importance, 4-and finally we will discuss the proposed solutions of these articles. Our own overall conclusion will come the last in part 5.

## 2. Blockchain fundamental technical overview

### *2.1. What is a blockchain*

As defined by Yaga et al. [29] a blockchain is a decentralized ledger technology (DL) that securely records transactions and eliminates the need of a central authority. This DL operates as a shared transactional database recording all changes applied to a transaction. The DL is accessible by all the members of the same network [24] and every member has the same version of the truth and can validate the authenticity of the transaction at any time.

To maintain the data integrity, transactions must go through a validation process before considered completed, this validation is done through the use of a pre-defined consensus mechanism [11]. This mechanism adds an additional level of trust between the participants and help to ensure the security of the DL. The DL is further protected by encrypted hash algorithms that secure each data block and the entire ledger [1].

In a simple term, blockchain is a decentralized ledger technology that uses a pre-defined consensus protocol, each block in the ledger represents a transaction and it is securely linked to the previous block forming the blockchain, the participants of this blockchain network (the nodes) use encryption to secure their transactions and protect the data in the ledger.

### *2.2. Different Types of blockchain*

According to Guegan [12], there are several varieties of blockchain network technology.

- Public blockchain: Also known as Permission-less blockchain, where anyone can become a node, participates in adding/validating blocks and reading/writing form and to the ledger. There is no central authority overseeing the network, making it open to everyone. Bitcoin is the obvious example of this is type of blockchain.
- Private blockchain: All nodes are interconnected in a private network where the members are known to each other. This particular blockchain network is reserved for authorized members only, as it is managed by a single entity. It is commonly employed for commercial applications.
- Consortium blockchain [28] is another variation of a permissioned blockchain, its managed by a group of organizations instead of a single entity. This type of blockchain provides a higher level of decentralization and security compared to the private one.
- Hybrid blockchain [4]: This type of blockchain is managed by a single entity or organization, with oversight provided by a public blockchain.

### *2.3. Benefit of blockchain*

Some of the major advantages of blockchain that are relevant to our study are as follow:

- Trust improvement: The members of the blockchain network are guaranteed to receive accurate data in a timely manner. The data is only shared internally and protected from outsiders, which leads to increased transparency and trust in the transactions.
- Security enhancement: Once the members of the blockchain network approve a transaction, it becomes immutable and recorded in a block permanently. No one, including the network administrator, can delete or modify a transaction. Furthermore, each member of the network holds a copy of the complete blockchain, which makes it highly unlikely for data to be altered [21].
- Greater Efficiency: Due to the shared DL among network members, the need for record reconciliation and waiting time is eliminated. Smart contracts, which are a major component of blockchain play a significant role in speeding up transactions.

## 3. Comparison between of the current claim adjudication process and a blockchain-based solution.

In this paragraph, and to set the stage for the review of the articles in our study, we will compare the current IT/manual system versus the proposed blockchain model to showcase the significant potential of using blockchain technology for health claim adjudication. This comparison demonstrate how blockchain can effectively replace the existing model and provide more efficient and secure claims adjudication process.

### *3.1. The current Model in place*

The current methods for adjudicating insurance claims varies from manual, semi-automated/hybrid, to fully automated processes. In the manual process model the entire adjudication process is done manually (submitting a printout form, manually filled in this form, agents review and process claim manually…), while in the semi-automated/hybrid model users (service providers/insurers) can submit claims using a web interface or an application while the other aspects of claim processing such as claim verifying and settlement are still performed manually. In contrast, the fully automated model the entire process: from claim submission, adjudication, payment settlement and auditing are all done thought a fully IT system using a dedicated network infrastructure, web interface portals, internal claim processing applications, batch files and a payment mechanism. The time it takes settle the payment to insurance companies/insurers in this fully automated system varies depending on the payment method used, ranging from the next business day for direct deposit to up to 15 days for checks. [27]

the diagram below in Fig. 1. illustrates the claim adjudication process for the fully automated model.

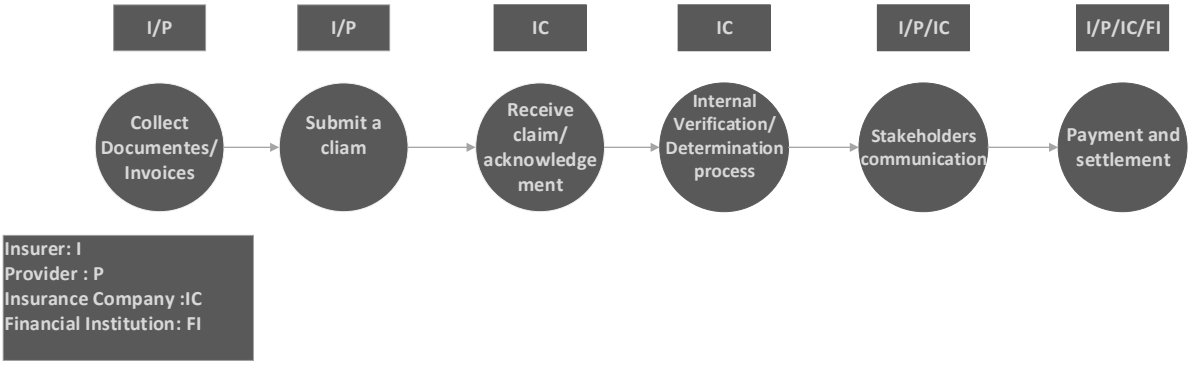



Fig. 1. fully automated claim adjudication model

*3.2 The Blockchain model: how Blockchain can be used for insurance claims:*

All members of the ecosystem are connected through a private Blockchain network. In this network, claims are processed and completed within minutes through a series of steps: the service provider (SP) shares insurer data with the insurance company (IC), which validates the insurer's identity and type of coverage. The SP then submits the claim to the IC, the insurer/patient approves the transaction for instant fraud prevention. The IC processes the claim and sends a settlement request to the financial institution (FI), which issues payment to the provider/insurer. The Blockchain technology used in this model allows all stakeholders to access the same data block transaction and update it instantly, using a predefined consensus protocol to approve any modifications. The content of each transaction is encrypted with a Private/Public key algorithm, and the data block is connected to the previous one to form the blockchain (Fig. 2). One significant advantage of using blockchain technology is the ability to facilitate real-time claim adjudication. [7].



Fig. 2. Blockchain claim adjudication model

## 4. Literature review

In the recent years, many studies tried to explore the opportunity to integrate blockchain into insurance. But due to the vast scope of the industry, each study tried to approach the insurance industry from a very specific angle and suggests a distinctive blockchain solution for. In this review we have selected only the most recent articles related to claim processing and adjudication in the health insurance, all other blockchain/insurance studies outside this scope were excluded; We have divided this study into four main titles. First, we have collected all the challenges that the traditional adjudication system is facing from these articles point of view. The following part is presenting the blockchain technical solution for each study. And since the fraud is a major issue that the insurance sector is facing we have dedicated the third paragraph to illustrate how each proposed solution addressed the fraud and what technology has been used to resolve it. the last part of this study is the technical discussion of the majority of these proposed solutions.

### *4.1. The Main Challenges facing the current claim processing systems*

In certain markets, where technology adoption is not yet widespread and claims are processed manually [2], there are several challenges that need to be overcome. These include complexity [9], [16], [24], limited accessibility to insurance services [24], bureaucracy [17], inefficiency [9], [17], [24], high administrative costs [2], security and privacy limitations due to the use of a centralized server model [23], and system abuse [19].

Other challenges that are common, even in countries that have implemented advanced IT solutions for claims adjudication are: The lack of transparency [9], [6], [19], [22], [25], high costs of the claim processing [2], [25] fraudulent claims [2], [6], [9], [11], [14], [16], [18], [19], [23], [24], delays in the claim handling process [2], [9], [25] difficulty in managing and verifying policyholder identities [2], inadequate security and privacy of insurance data [2], [5], [10], [13], [15], [17], [19], [22], [24], [26] ,[30], lack of trust among the stakeholders [6], [9], [15], [16], [22], [30], the high risk of errors in insurance claims processing [6], the need of a secure and efficient data management platform [6], [17], operational regulatory [6], [26], and reputational risks [26] and the increasing complexity of insurance regulations [6].

### *4.2. The proposed solutions*

Several blockchain technologies solutions have been proposed in these articles to provide a secure, transparent, and decentralized system for insurance claim processing. These solutions have the ability to disrupt the insurance industry by automating and streamlining the claims process, reducing processing time, and preventing fraudulent activity.

Some solution involves the use of smart contracts and IoT devices [24], [26] to automate the claims process resulting in faster refunds and cost savings for insurance companies. Shared ledgers [1], [20] allow all parties involved to process claims simultaneously, further reducing processing time.

Another proposed solution is a blockchain framework for insurance processes that includes several components such as smart contracts [2], [3], [5], [13], [18], [19] triggering the claims process [9] and payment [13], consensus mechanisms [30] that securely and immutably store the data related to policies, claims, and other insurance-related transactions [18]. These components work together to enable secure and efficient insurance processes [2], establish a transparent and tamper-proof system [16], [19] for health claim processing, some solutions [19] integrated a framework comprises of four primary components, and it aims to provide a secure, transparent, and decentralized healthcare system, which is accessible to all stakeholders. Other introduced a machine learning algorithms [9] to: identify potential fraudulent claims, store all insurance claim-related data, including customer information, insurance policies, and claim records [9] and improve the quality of information, user satisfaction, and prevent fraud.

ClaimChain blockchain solution [22] is a secure blockchain platform presented as well to handle insurance claims processing. It can streamline the claims process and prevent fraudulent activity, providing a thorough evaluation of the security and privacy. The B-SAHIC [16] system is a blockchain-based secured and automated health insurance claim processing system where the authors propose a private blockchain network that all the

stakeholders can participate in, smart contracts automate the entire process, and a mechanism for fraud detection and prevention were also included in this framework.

A scheme that uses a combination of smart contracts, encryption, and zero-knowledge proof technology [30] ensures the privacy and security of the insurance claim process was another proposed solution. This scheme also includes a ranking system that assigns a reputation score to each user based on their behaviour in the system, preventing fraud by detecting suspicious behavior.

A specific solution integrated smart contract and the use of Hyperledger for fraud detection combining with a decision-making map that includes multiple factors like consensus protocol, the type of the blockchain, privacy, security and scalability. This same solution involves the use of weights assigned to each factor, the platform with the highest score will be the most suitable, this how the business owner and the developers can stay informed when selecting the blockchain platform for their applications [14].

A blockchain-based healthcare system using the integration of UML diagrams to design the system and DApps offered by the Vexanium Ecosystem [17] to simulate claims between patients, doctors, and insurance companies was another solution used to authenticate and record medical records on a blockchain network.

### *4.3. How these studies addressed the fraud problem?*

One of the increasing and common challenges that most of the insurance companies are facing nowadays is the fraudulent claims, which can result in significant financial losses. The estimated fraud loss associated with fraudulent claims is beyond imagination, in 2019 the authorises were able to recover $2.6 billion in the US alone from fraudulent claims [19]. About 20 to 30% of insurance claims are suspected of fraud according to the calculations of the International Association of Insurance Supervisors (IAIS) [9] This is why, in this study we have dedicated a separate paragraph to discuss how the articles of this literature review addressed the issue utilizing blockchain technology.

Blockchain system would enable real-time transaction [3], [15], [24] and simultaneous processing [24]. The use of oracles [24] integrated within the blockchain solution would initiate the claim filing process automatically, increasing accuracy with less human intervention [24] and increase the potential to cut down on all activities related to fraud is high, as every transaction would be verified and recorded on the secure, immutable and tamper-proof blockchain ledger [3], [9], [10], [19], [24].

A lot of studies discussed in this review have integrated Smart contract in their solutions [1], [2], [3], [9], [14], [22], because it can automatically verify claims based on pre-defined conditions [2], it can also reduce the risk of fraudulent claims [6], [16], [19], eliminate the need for intermediaries [15], [16], [22], improve the transparency [3], [6], [9], [10], [30], [21] and have the characteristics of claims self-executing [2], [15], [9], [10], [24], where only authorized parties involved in the insurance process can have access to the same information [2], [6], [13], [22], which can help prevent duplicate [2] or fraudulent claims [2], [9], [15], [16], [18], [27], [30]. The incorporation of Biometric data [13] into the blockchain has increased security and prevented fraud. Zero-knowledge proof [30] was another approach used to achieve anonymously. Multi-party consensus mechanism scheme was included into one of the studies to improve the accuracy [30], [24] and reliability of claim adjudication [30] that reduces the likelihood of fraudulent claims, a similar variant to the multi-party consensus is the data security and privacy maintenance and multi-party data alliance among the stakeholder help to the transparency of the system for verifying the service provider and the patient at the same time [23]. Finally, some solution presented the use of blockchain and the Internet of Things (IoT) [26] to enhance security, auditability, privacy, and accountability and protected against financial losses due to fraud.

### *4.4. Discussion of the proposed solutions*

While the articles of this study provide a promising blockchain solution for the claim processing systems; Some of the proposed solutions need more: [16] research and development, [5], [16] critical evaluation and [25] further testing in real-world settings. And lack: [5], [17] scalability, [2] interoperability and [18] strong technical expertise. Other misses a detailed coverage on the [5] costs and risks associated with the implementation and regulatory compliance to uplift the system to a national level.

Other solutions [24] only focus on the benefits of the technology and are relatively narrow in scope or [14], [17] does not provide a security and privacy detailed analysis. Similarly, some solutions [2], [5], [18], [26] do not discuss the potential limitations and [2] lack concrete examples of implementation. For others [9], [16], [18], the empirical evidence of the effectiveness of blockchain in addressing the challenges is missing or [13], [16] it has not been fully addressed to determine the feasibility and the potential challenges associated with the implementation of a such solutions.

To address some of the drawbacks discussed earlier, several steps can be taken like: [16] introducing more insurance companies and conducting in-depth outcome tests would be more suitable than focusing solely on one company in the model. [2], [18], [25] providing a well-defined technical guideline for developing the proposed blockchain application with comprehensive technical coverage is necessary to illustrate the full potential of Blockchain technology and its advantages. [17] contributors must provide more detailed explanations to address the technical issues related to claim fraud and processing. [24] shifting the discussion from the policy level to the transactional claim level would provide a more practical and focused solution. Although the study addressed critical topics such as medical data management and claim processing [6], more detailed separate studies may be necessary to illustrate how Blockchain can provide alternative solutions for existing infrastructure. [19] implementing the proposed platform requires full cooperation from existing stakeholders, and its implementation will be costly and time-consuming due to the need for a full modification of existing infrastructure. Despite blockchain's main purpose being to eliminate the presence of a central authority and shift towards a decentralized environment, the study emphasizes the need for an essential central entity, the CMS “MS: Centres for Medicare and Medicaid Services”, to set smart contract rules, data, permission, network structure, and consensus parameters. Lastly, [22] integrating the massive existing National Insurance Crime Bureau "NICB" database with the ClaimChain and keeping it up to date is a significant challenge considering the substantial differences between existing and suggested infrastructures.

In the other hand some researchers have provided a complete overview that could be considered as a good start-up point toward the implementation of a full blockchain solution in the real word.

For example, in [19], the solution proposed that incorporated consortium blockchain with smart contracts presents a promising solution and has a real potential to make a significant contribution to the healthcare industry by providing a more secure, transparent, and accessible healthcare system but at the same time this specific solution found to be costly at the same time. Other system design [22] proved to be secure, efficient, and accessible to all participants in the insurance ecosystem.

## 5. Conclusion

The studies presented in this paper highlight the potential of blockchain technology to completely transfer the insurance and healthcare industries and demonstrated a suitable alternative to the existing adjudication model. They range from a broad discussion about the benefits of blockchain to a detailed technical solution. It's evident that blockchain has the ability to improve claims processing, increase data security and confidentiality through encrypted transactions, and reduce the cost of implementation and maintenance compared to other systems. Some studies even showcased the possibility of combining blockchain with AI and come up with new and innovative solutions. The use of blockchain and smart contracts can lead to a more efficient, accurate, and cost-effective claims process by ensuring transparency and security throughout the process.

It is worth it to mention as well that these proposed solutions come from various countries and economies, and address the topic in multiple technical and academic ways. However, since the insurance and healthcare industries are heavily regulated and a technical study is not enough alone to promote the implementation of such a solution. Any proposed blockchain model should take into account the specific regulations of each country and tailor the blockchain solution around to meet those requirements.

This is why we refrained to include our study into this review, because it addresses the blockchain model from the similar way most of the other studies did. We might consider to build our blockchain solution around the regulations of a specific market (The Canadian market in this instance), define its rules and regulation, and then building the blockchain solution accordingly. This could be our next more specialized academic research.

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